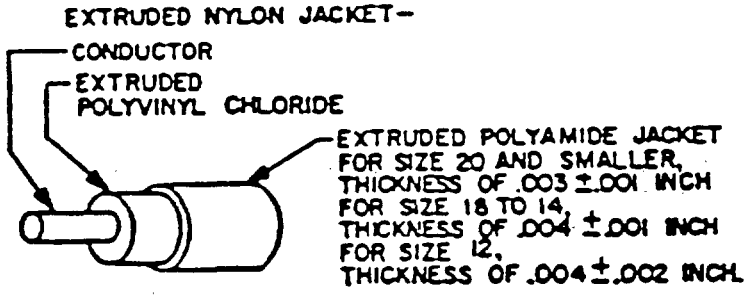
INCH-POUND
MIL-W-16878/18A(NAVY)
11 September 1992
SUPERSEDING
MIL-W-16878/18(NAVY)
10 August 1981

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, POLYVINYL CHLORIDE (PVC) INSULATED, 105°C, 1000 VOLTS, POLYAMIDE JACKET (NOT FOR NAVY SHIPBOARD USE)

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-16878.



(NOT FOR NAVY SHIPBOARD USE)

FIGURE 1. Construction.

AMSC N/A FSC 6145

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TABLE I. Construction details.

(See footnotes at end of table.)

PIN 3/	Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material 1/ 2/	Coating		Min	Max
M16878/18-BDA*	26	1 X 26	Copper	Tin	0.0159	0.050	0.062
M16878/18-DDA*	26	1 X 26	H.S.C.A.	Silver	.0159	.050	.062
M16878/18-CDA*	26	1 X 26	C.C. steel	Tin	.0159	.050	.062
M16878/18-BDB*	26	7 X 34	Copper	Tin	.019	.053	.065
M16878/18-DDB*	26	7 X 34	H.S.C.A.	Silver	.019	.053	.065
M16878/18-BDE*	26	19 X 38	Copper	Tin	.021	.053	.065
M16878/18-DDE*	26	19 X 38	H.S.C.A.	Silver	.021	.053	.065
M16878/18-BEA*	24	1 X 24	Copper	Tin	.0201	.054	.066
M16878/18-DEA*	24	1 X 24	H.S.C.A.	Silver	.0201	.054	.066
M16878/18-CEA*	24	1 X 24	C.C. steel	Tin	.0201	.054	.066
M16878/18-BEB*	24	7 X 32	Copper	Tin	.024	.058	.070
M16878/18-DEB*	24	7 X 32	H.S.C.A.	Silver	.024	.058	.070
M16878/18-BEE*	24	19 X 36	Copper	Tin	.026	.058	.070
M16878/18-DEE*	24	19 X 36	H.S.C.A.	Silver	.026	.058	.070
M16878/18-BFA*	22	1 X 22	Copper	Tin	.0254	.059	.072
M16878/18-DFA*	22	1 X 22	H.S.C.A.	Silver	.0254	.059	.072
M16878/18-CFA*	22	1 X 22	C.C. steel	Tin	.0254	.059	.072
M16878/18-BFB*	22	7 X 30	Copper	Tin	.030	.064	.076
M16878/18-DFB*	22	7 X 30	H.S.C.A.	Silver	.030	.064	.076
M16878/18-BFE*	22	19 X 34	Copper	Tin	.032	.064	.076
M16878/18-DFE*	22	19 X 34	H.S.C.A.	Silver	.032	.064	.076
M16878/18-BGA*	20	1 X 20	Copper	Tin	.0320	.066	.078
M16878/18-DGA*	20	1 X 20	H.S.C.A.	Silver	.0320	.066	.078
M16878/18-BGB*	20	7 X 28	Copper	Tin	.038	.072	.084
M16878/18-DGB*	20	7 X 28	H.S.C.A.	Silver	.038	.072	.084
M16878/18-BGC*	20	10 X 30	Copper	Tin	.038	.072	.084
M16878/18-DGC*	20	10 X 30	H.S.C.A.	Silver	.038	.072	.084
M16878/18-BGE*	20	19 X 32	Copper	Tin	.041	.072	.084
M16878/18-DGE*	20	19 X 32	H.S.C.A.	Silver	.041	.072	.084
M16878/18-BHA*	18	1 X 18	Copper	Tin	.0403	.075	.088
M16878/18-BHB*	18	7 X 26	Copper	Tin	.049	.084	.096
M16878/18-BHE*	18	19 X 30	Copper	Tin	.051	.084	.096
M16878/18-BJA*	16	1 X 16	Copper	Tin	.0508	.085	.098
M16878/18-BJE*	16	19 X 29	Copper	Tin	.059	.094	.106
M16878/18-BJF*	16	26 X 30	Copper	Tin	.062	.097	.109
M16878/18-BKA*	14	1 X 14	Copper	Tin	.0641	.099	.111
M16878/18-BKE*	14	19 X 27	Copper	Tin	.072	.107	.119
M16878/18-BKH*	14	41 X 30	Copper	Tin	.080	.115	.127

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PIN 3/	Wire size	Strand-ing	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material 1/ 2/	Coating		Min	Max
M16878/18-BLA*	12	1 X 12	Copper	Tin	0.0808	0.116	0.131
M16878/18-BLE*	12	19 X 25	Copper	Tin	.091	.127	.141
M16878/18-BLG*	12	37 X 28	Copper	Tin	.089	.125	.139
M16878/18-BLJ*	12	65 X 30	Copper	Tin	.098	.139	.148

1/ H.S.C.A. stands for high-strength copper alloy.
2/ C.C. stands for copper-clad.
3/ PIN stands for part or identifying number.

ADDITIONAL REQUIREMENTS:

Visual and mechanical examination: Required.
Spark test: 5.0 kV.
Impulse dielectric test: 10.0 kV.
Dielectric withstanding voltage: 3.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$.

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C.
K = 2000.
D = Maximum average diameter of finished wire.
d = Conductor diameter.

Cold bend: Condition 4 hours at minus 54 ± 1°C (see table II).

TABLE II. Cold bend mandrel sizes.

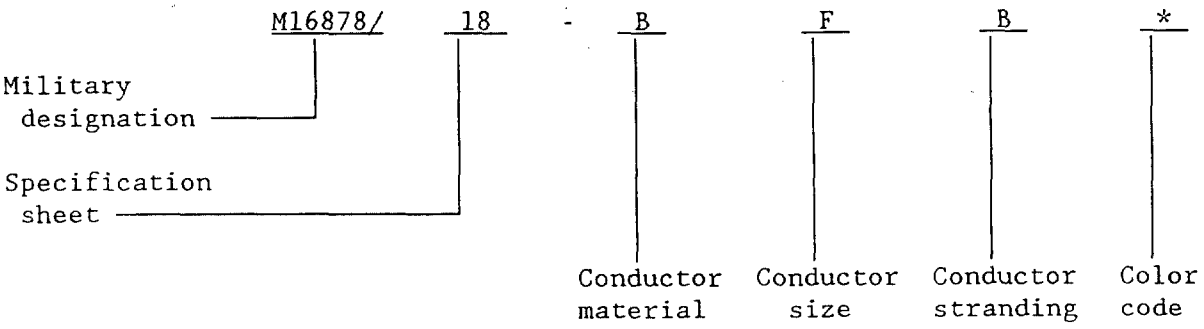
Wire size	Cold bend mandrel (maximum diameter in inches)
26 through 16	2
14, 12	3

Concentricity: 70 percent (minimum).
Surface resistance: Not required.
Wrap back: Not required.
Heat resistance: Condition at 150°C. Cracking of the jacket shall constitute a failure.
Flammability: Meets requirements of UL VW1 Flame Test.
Toxicity: Meets requirements of concentration levels specified in MIL-E-917.
Shrinkage: Required.

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Heat aging: 25 percent change (maximum) in 96 hours at 136°C.
Insulation tensile strength: 1800 pounds force per square inch (minimum).
Insulation elongation: 100 percent (minimum).
Marking and stripe durability: Required.
Fungus resistance: Required.
Maximum dc resistance of finished wire: See table I of MIL-W-16878.
Identification of product: Required for wire sizes 22 AWG and larger.

PIN example (see MIL-W-16878):



Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6145-N330-17)

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